Elderly Care Management System

Comprehensive High-Fidelity UI/UX Prototype, Design System Tokens, and Interface Architecture for Geriatric Home Healthcare

ACADEMIC SUBMISSION PROFILE

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Project Title: Elderly Care Management System
Design Tooling: Figma / Canva Vector Grade Architecture
Date: October 2026

DESIGN SYSTEM OBJECTIVES

- **Senior-Centric Readability:** WCAG AAA contrast, minimum 14pt body text, and prominent visual cues.
- **Zero-Error Medication Safety:** Color-coded dosage status with time stamps and audio-visual cues.
- **Caregiver Accountability:** Structured daily handoff logs, vitals verification, and instant emergency dispatch.

SEQUENCED INTERFACE SPECIFICATION (1 SCREEN PER PAGE)

1. Design System Tokens & Style Guide
2. Screen Flow & Information Architecture
3. Screen 1: Role-Based Authentication
4. Screen 2: Central Family Dashboard
5. Screen 3: Elderly Profile & Health Record
6. Screen 4: Medicine Schedule & Tracker
7. Screen 5: Add Medication Form Modal
8. Screen 6: Doctor Appointment Tracking
9. Screen 7: Daily Care Routine Checklist
10. Screen 8: Caregiver Shift Handoff Log
11. Screen 9: Instant Emergency SOS Portal

EC 01. Design System & Design Tokens

Color Palette, Typography Scale, UI Components, and Accessibility Standards

Design System v2.0 | Figma Tokens

Core Medical Color Tokens

Teal Primary
#0D9488 • Brand & CTAs

Deep Slate
#0F172A • High Contrast Text

Amber Alert
#F59E0B • Med Reminders

Emergency Red
#E11D48 • SOS & Paramedics

Typography Hierarchy

Heading 1 (24px) — Section Titles

Heading 2 (18px) — Card Headers

Subheading (14px) — Data Labels

Body Regular (13px) — High legibility medical text

Interactive Badges & Buttons

✓ Dose Completed | ⏳ Due in 25m | 🚨 Critical SOS | 🩺 Doctor Visit

Primary Action | **Secondary Option**

Emergency Dial

Geriatric & Elderly-Friendly UX Guidelines

- **Touch Targets:** Buttons and clickable tiles minimum 44px height.
- **WCAG AAA Contrast:** Minimum 7:1 contrast ratio on all key text.
- **Dual Encoding:** Information conveyed using both text + color icons.

DESIGN PHILOSOPHY

A human-centered system designed specifically to bridge the communication gap between busy family members, non-technical elderly users, and home healthcare aides.

DESIGN TOKENS

#0D9488 Brand | #0F172A Base | #F8FAFC Canvas | #E11D48 Urgent | Inter / Plus Jakarta | 8px Radius Grid

KEY STANDARDS

- Card-based modular layouts.
- Real-time timestamp verification.
- Optimized for desktop, laptop & tablet screens.

02. Screen Flow & Information Architecture

End-to-End User Journey from Authentication to Daily Operations and Emergency Protocol

Architecture | User Journey



ARCHITECTURE STRATEGY

The system utilizes a **Hub-and-Spoke Topology**. The Dashboard functions as the persistent anchor, ensuring non-technical users never get lost in deep sub-pages.

SCREEN SEQUENCE

- **Step 1:** Role-based entry
- **Step 2:** Main state overview
- **Step 3–7:** Core healthcare ops
- **Step 8:** Care staff coordination
- **Step 9:** Instant emergency rescue

EC

03. Screen 1: Role-Based Authentication Portal

Secure login with instantaneous dual-role switching between Family Members and Caregivers

Screen 1 of 9 | Auth & Roles

https://elderlycare.portal/login

SSL 256-bit Encrypted

Elderly Care Portal

Dedicated Family & Caregiver Monitoring System

Family Member | Caregiver Staff

Email Address or Phone Number

tariq.rahman@caremail.com

Password — Forgot Password?

••••••••••••••

Keep me securely logged in on this tablet

Sign In to Care Dashboard ➔

Need immediate emergency dispatch?

Call 24/7 Ambulance: 999 / 10616

SCREEN PURPOSE

Guarantees safe role-isolated login. Family members access billing and prescriptions, while caregivers access clinical logs and shift reports.

UX DETAILS

- **Instant Role Switch:** Single-click toggle between Family and Caregiver profiles.
- **Large Tap Targets:** Designed for adult children and senior nurses on mobile/iPad.
- **Persistent Emergency Line:** Critical phone numbers visible even before login.

FIGMA TOKENS

#0D9488 Primary | 12px Card Radius | Shadow-LG

04. Screen 2: Central Family Member Dashboard

Comprehensive status command center: vitals summary, upcoming medicine alert, daily routine & SOS

Screen 2 of 9 | Core Command Center

https://elderlycare.portal/dashboard • System Live

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

User: Tariq (Son)
Family Admin

Active Care Profile: Abdul Rahman Chowdhury (76 Yrs) | Vitals Stable | ONE-TOUCH SOS

Next Scheduled Dose: Atorvastatin 20mg (1 Tab) Due after lunch in 25 mins — ✓ Mark Taken

BLOOD PRESSURE	TODAY'S MEDICINES	NEXT APPOINTMENT	CAREGIVER ON DUTY
124 / 82 mmHg	3 of 5 Taken	Tomorrow 10:30 AM	Nasreen Akhtar
Normal Reading	60% Complete	Dr. Hasan (Cardio)	Active Shift

Today's Care Schedule (Sunday, Oct 4) — View Full Routine ➔

- **08:00 AM** • Morning Walk & Breathing — ✓ Completed
- **08:45 AM** • Breakfast & Morning Meds — ✓ Completed
- **01:30 PM** • Healthy Lunch + Heart Dose — Due Soon
- **05:00 PM** • Evening Sugar Check — Scheduled

Caregiver Live Observation Notes — + Add Note

Logged by Nasreen (11:20 AM):
"Patient ate full breakfast with great appetite. Blood sugar checked at 110 mg/dL. Finished 15 mins garden stroll safely."

Upcoming Doctor Preparation:
Echo reports from August kept ready in folder for tomorrow's visit with Dr. Hasan at Square Hospital.

SCREEN OBJECTIVE

Provide family members complete peace of mind in under 5 seconds. Immediate overview of vitals, adherence percentage, and live caregiver shift notes.

UX COMPONENTS

- **Amber Alert Banner:** Highlights the immediate next medication.
- **Stat Cards:** Real-time BP, pill count, and duty nurse status.
- **Instant SOS:** Persistent Red trigger for immediate help.

DATA HANDLED

Vitals Stream | Adherence % | Shift Status

05. Screen 3: Elderly Profile & Medical History

Comprehensive electronic health record: personal demographics, chronic conditions, and physician contacts

Screen 3 of 9 | EHR & Demographics

https://elderlycare.portal/profile/senior-01

Edit Records

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Record ID: EC-2026-881

Elderly Health Profile & Emergency Summary

⚠ Penicillin Allergy



Abdul Rahman Chowdhury

Blood Group: B+ (Positive)

Age: **76 Years** • Male • Born: 14 Aug 1950 • Weight: 68 kg • Height: 5' 7"

Residential Address: House 42, Road 11, Dhanmondi R/A, Dhaka-1209

Mobility: Walking Cane

Insurance: Active (Pragati Life)

Chronic Conditions

- Type-2 Diabetes — Since 2012
- Hypertension (High BP) — Since 2015
- Coronary Stent — Placed 2021
- Mild Osteoarthritis — Left Knee

Primary Doctors

Dr. K. M. Hasan
Cardiologist, Square Hospital
Call

Dr. Nusrat Jahan
Endocrinologist, Labaid
Call

Care Directives & Diet

- **Diet:** Strict low-sodium, diabetic-friendly meals (under 1500 kcal).
- **Hydration:** 200ml fluid intake prompt every 2 hours.
- **Fall Prevention:** Non-slip bathroom mats & walking cane assistance mandatory.

SCREEN PURPOSE

A single-pane medical repository for attending clinicians and rotating caregivers, ensuring critical allergies and pre-existing conditions are never overlooked.

UI FEATURES

- **Critical Allergy Tag:** Penicillin alert highlighted in vivid red.
- **Direct Calling:** Fast dial triggers for primary treating physicians.
- **Mobility Flag:** Prepares caregivers for physical support needs.

DATA ARCHITECTURE

Demographics | Allergy Protocol | Physician Links

06. Screen 4: Medicine Schedule & Tracker

Chronological prescription tracker organized by Morning, Afternoon, Evening, and Bedtime slots

Screen 4 of 9 | Medication Safety

https://elderlycare.portal/medications

+ Add Medication

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Daily Dose Target: 5 Pills

Today's Schedule (Oct 4) | All Prescriptions (8) | Refill Alerts (2) | 3 of 5 Taken

MORNING ROUTINE (08:00 AM – 09:00 AM)

✓ All Taken (Verified by Nurse Nasreen)

Metformin 500mg (1 Tablet) — Taken 08:35 AM
With Breakfast • Dr. Nusrat

Amlodipine 5mg (1 Tablet) — Taken 08:35 AM
After Breakfast • Dr. Hasan

AFTERNOON ROUTINE (01:30 PM – 02:00 PM)

⏳ Due in 25 Minutes

Atorvastatin Lipitor 20mg (1 Tablet - Cholesterol)
Take immediately following lunch with water • Prescribed by Dr. Hasan

✓ Mark Taken | Snooze 15m

NIGHT & BEDTIME ROUTINE (09:30 PM – 10:00 PM)

2 Doses Scheduled

Metformin 500mg (1 Tablet) — Upcoming
After Dinner

Calcium D3 (1 Tablet) — Upcoming
With Warm Milk

SCREEN PURPOSE

Eliminates missed or doubled medication doses. Real-time logging creates a verified digital audit trail with nurse verification.

SAFETY CONTROLS

- **Chronological Sorting:** Clear time buckets prevent confusion.
- **One-Click Confirmation:** Fast "Mark Taken" button updates family dashboard instantly.
- **Refill Tracking:** Automatic counter notifies when 5 pills remain.

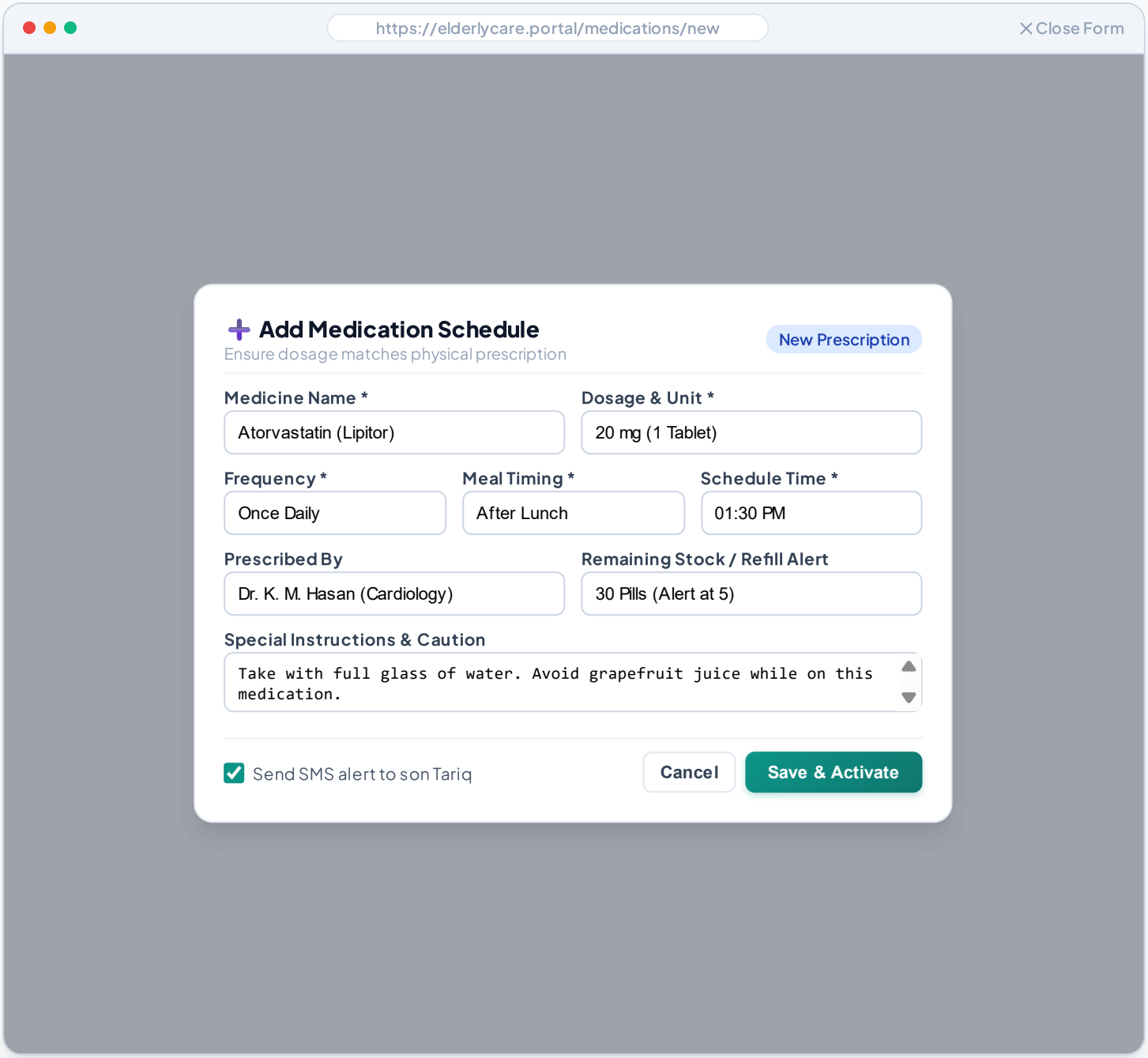
AUDIT TRAIL

Timestamped | Nurse ID | Missed Alert

07. Screen 5: Add / Edit Medication Form

Prescription entry form with dose validation, meal-timing association, and low-stock refill triggers

Screen 5 of 9 | Prescription Input



FORM VALIDATION

Medical safety constraints prevent accidental overdose entries or uncalibrated frequencies.

KEY FIELDS

- **Prescribing Doctor:** Links the dose directly to clinical consultations.
- **Meal Timing:** Before meal / with food / after meal selectors.
- **Inventory Counter:** Automatically computes days until refill needed.

MODAL BEHAVIOR

Backdrop Blur | Required Validation | SMS Hook

08. Screen 6: Doctor Appointment Tracking

Medical appointment calendar, physician directory, consultation reports, and family escort details

Screen 6 of 9 | Clinical Scheduling

https://elderlycare.portal/appointments

+ Book Visit

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Next: Tomorrow 10:30 AM

Doctor Consultation Schedule & History

SMS Alert: 24h & 2h Prior

NEXT SCHEDULED CONSULTATION

Dr. K. M. Hasan — Senior Cardiologist

Square Hospital, Panthapath • **Tomorrow (Monday, Oct 5) at 10:30 AM** (Room 314)

Clinical Goal: 3-Month Echo & ECG Review • Escort: Tariq (Son) • Transport Arranged

View Hospital Map

Notification Active

Future Scheduled Visits

Dr. Nusrat Jahan (Diabetes)

Labaid Diagnostic, Dhanmondi

Oct 18, 2026 • 05:00 PM

(Fasting Blood Sugar)

Reschedule

Dr. Shahed Ali (Orthopedic)

Popular Hospital, Shantinagar

Nov 02, 2026 • 11:00 AM (Knee Joint Mobility)

Reschedule

Past Consultation Records & RX

Routine Cardiology Follow-up

Sep 04, 2026 • Dr. K. M. Hasan

(BP: 120/80)

View RX

Ophthalmology Vision Assessment

Aug 12, 2026 • Vision stable, no cataract surgery

View RX

SCREEN PURPOSE

Centralizes all doctor appointments so family members can coordinate transport, tests, and family escorts well in advance.

FEATURE HIGHLIGHTS

- **Escort Designation:** Clearly designates which son/daughter will attend.
- **Digital Prescriptions:** Instant access to scanned prescriptions from past visits.
- **Map Integration:** Direct clinic navigation directions.

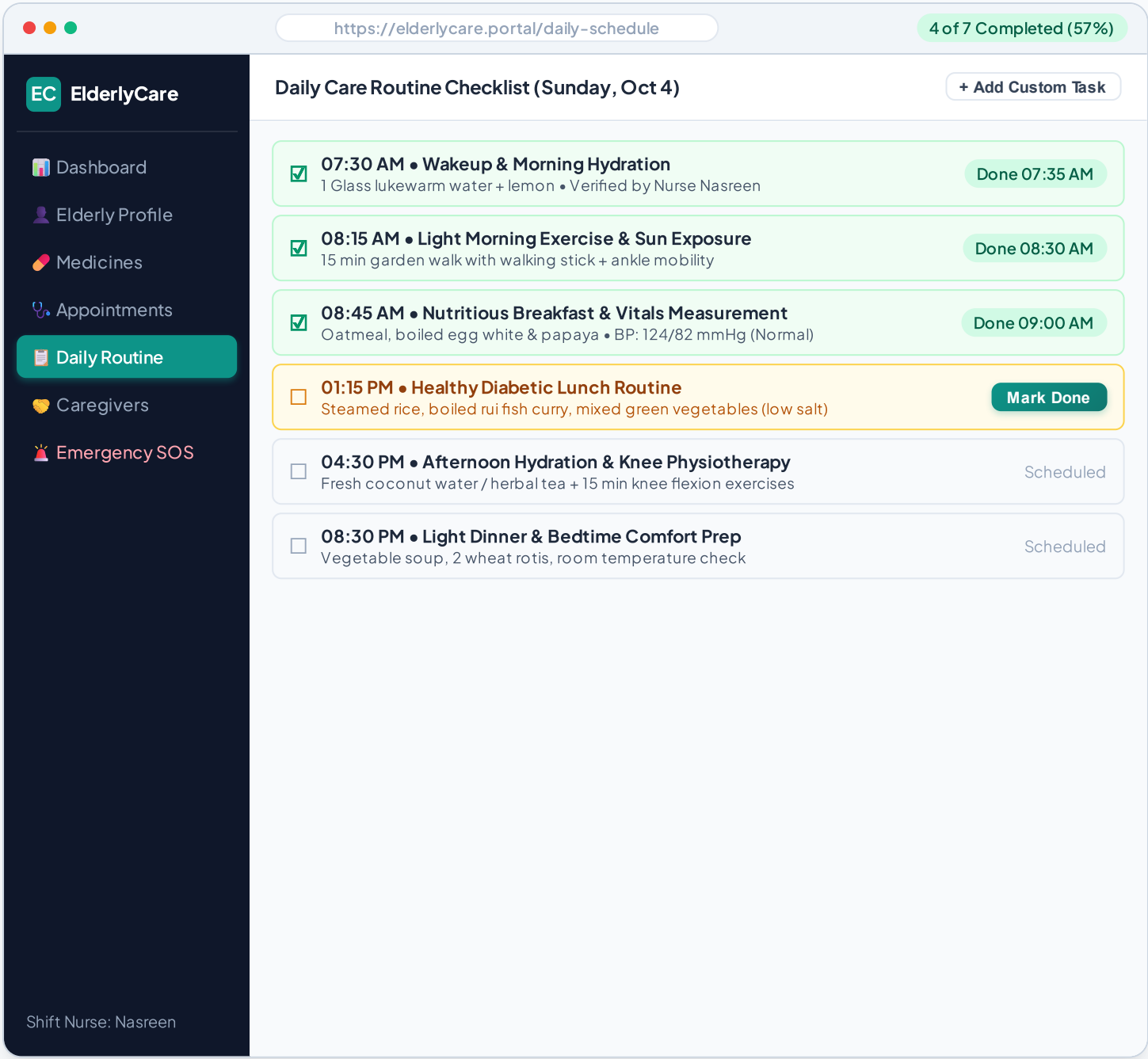
SYSTEM INTEGRATIONS

Google Calendar | SMS Reminders | PDF RX Archive

09. Screen 7: Daily Care Schedule & Activities

Holistic elderly lifestyle checklist: meals, hydration checks, light mobility, and vitals recording

Screen 7 of 9 | Daily Routine Checklist



HOLISTIC HEALTH FOCUS

Comprehensive care requires nutrition, hydration, and mobility, not just pills. This screen turns clinical directives into an achievable daily routine.

KEY FEATURES

- **Digital Checkbox:** Tap to toggle status with instantaneous timestamp.
- **Meal Composition:** Explicitly states low-salt / diabetic instructions.
- **Caregiver Verification:** Prevents missed nutrition or hydration gaps.

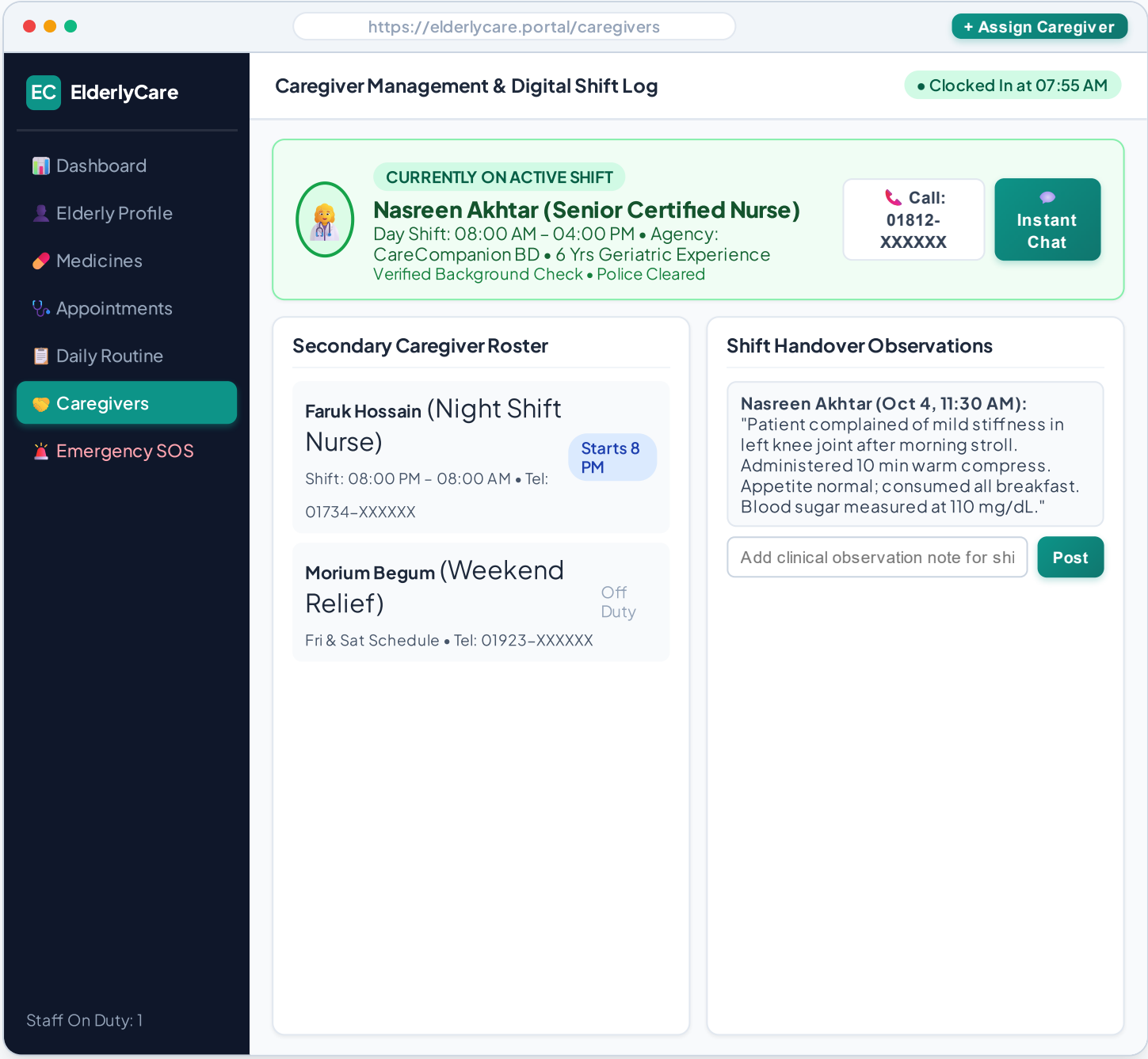
CATEGORIES

Nutrition | Vitals | Hydration | Mobility

10. Screen 8: Caregiver Information & Shift Handoff Log

Caregiver directory, verified credentials, shift clock-in, duty checklist, and observation exchange

Screen 8 of 9 | Caregiver Coordination



SCREEN PURPOSE

Facilitates smooth clinical handoffs between morning and night caregivers while keeping family members informed of daily behavioral changes.

KEY FEATURES

- **Direct Connectivity:** Family can phone or WhatsApp the caregiver with 1 tap.
- **Verified Staff Credentials:** Displays nurse training and police clearance.
- **Observations Log:** Real-time logging of physical discomfort or symptoms.

ACCOUNTABILITY

GPS Geofence | Clock In/Out | Handoff Notes

11. Screen 9: Instant Emergency SOS & Paramedic Brief

Rapid one-touch emergency dispatch, hospital ER direct routing, and immediate medical summary

Critical Safety Protocol | Screen 9 of 9

https://elderlycare.portal/emergency-sos

RAPID RESPONSE ACTIVE

SOS ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Hotline: 999 / 10616

IMMEDIATE EMERGENCY BROADCAST CONSOLE

TRIGGER BROADCAST SMS TO ALL FAMILY

National Ambulance

999 / 199

CALL AMBULANCE NOW

Square Hospital ER

10616 / 02-8144400

CALL ER DESK NOW

Dr. Hasan (Cardio)

01713-XXXXXX

CALL DOCTOR DIRECT

Family Emergency Contacts (Priority Order)

Tariq Rahman

Son • Priority 1

Lives 10 mins away • Tel: 01711-XXXXXX

Call 1st

Farzana Chowdhury

Daughter • Priority 2

Tel: 01819-XXXXXX • Alternative Contact

Call 2nd

Paramedic Critical Health Brief (On Arrival)

- **Patient:** Abdul Rahman Chowdhury (Age: 76 Years)
- **Blood Group:** B Positive (B+)
- **CRITICAL ALLERGY:** PENICILLIN (Causes Anaphylaxis)
- **Key Conditions:** Type 2 Diabetes, Hypertension, Coronary Stent (2021)
- **Home GPS Coordinates:** 23.7461° N, 90.3742° E (Dhanmondi Road 11)

CRITICAL LIFE-SAFETY PROTOCOL

In medical crises, every second counts. This screen provides massive single-touch speed dial triggers and broadcasts the senior's exact home location and vital history instantly.

EMERGENCY CAPABILITIES

- **One-Tap Speed Dial:** Direct links to National Ambulance and Hospital ER desk.
- **Paramedic Summary:** Prevents lethal drug reactions (Penicillin allergy clearly flagged).
- **Automated Family Ping:** Sends distress SMS with GPS pin to all relatives.

EMERGENCY TOKENS

#E11D48 Vivid Red | Speed Dial | GPS Lat/Long